Portable Measurement Arm Joint Design

*Interesting topic, the groundwork has already been done simple scope.

Reference	:	KS4
Supervisor	:	Prof Kristiaan Schreve
Can be completed in first semester	:	No
Students allowed for this topic	:	2
Student already assigned	:	None
Direction	:	Mechatronic
Required elective	:	None

Description:
Portable measurement arms are versatile tools to take 3D coordinate measurements. They normally offer six degree of freedom movement and the design can be adapted to various measurement envelopes. See https://www.wdhearn.co.za/product/portable-measuring-arms for a selection of typical measurement arms. The arm typically has four revolute joints. The joints are not motorised as a robot arm is. An operator must move the arm manually, typically by grabbing onto the last segment of the arm. The arms offer accuracies in the order to 5 to 50 microns, often depending on the size of the arm.
A key feature of the arms is that their revolute joints provide sufficient holding torque to keep the arm in position after the operator removed their hand from the arm. It would be a very impractical device otherwise!
They joints are also very important in terms of the system accuracy. The run-out tolerances on the joint must be very tight. An encoder is also incorporated in the joint to measure the angular displacement. Finally, there are cables running through all the joints to take the signals to the control unit. Obviously, the joint must be compact and light weight. The holding torque capability must not degrade over time.
A quick internet search did not provide any commercially available joints that meet these requirements. Friction hinges are a low cost, very inaccurate, option that can be used as some inspiration for the design. One website suggests damping grease like Nyogel, but it is doubtful that this will provide enough holding torque.
In this project, a mathematical model of the error propagation of a typical measurement arm must be used to determine the required tolerances for the joint. The model was developed by Leon Hall, a previous skripsie student, in 2009. Using this specification, the joint must be designed and test. Testing is on a single joint only. Building the full arm is outside the scope of this project.

Key modules (Scientific and Engineering Knowledge):
Design (E) (314) | Machine Design A (314)

Have been exposed to module

REPORT CONTENTS

- 1 **Background**: Start from the big picture, gradually narrow focus down to this project and where this report fits in
- **Aim & objectives**
  - implementing a position-holding mechanism on the joint of measurement arm w/o jeopardising its accuracy
  - Aim is usually one sentence stating the overarching aim of the project
  - Objectives of the project are usually 3 bullet points each
    - maintain or improve accuracy in the working volume
    - to make the joint design modular & applicable to other types PMCAs
    - To propose a calibration and/or testing method for the arm given a new joint design
- **Scope**: Give limitations of the project to clarify the objectives
- **Motivation**: Why is this project worthwhile
- **Literature review** → LIVING DOCUMENT → Must be updated regulated
- **Research methodology** → Describe research methodology (<1 page)
- **Workplan** → Discuss project steps. Each step should have a deliverable. (1-2 pages)
- **Resource requirements** → Linking back to the workplan, describe (list) the resources that will be needed to execute the project. (≤1 pages)

  Eg. Labs, funding, equipment, materials, software.
- **Expected results**: Describe the expected results (≤1 page)
- **ECSA strategy must included** → Appendix